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Structural safety risk prediction method for terminal building steel roof construction considering spatial and temporal variations

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ABSTRACT

In the field of aviation, terminal building renovation and expansion projects frequently involve complex construction processes and the structure is in a non-static state during the construction process, resulting in significant safety risks accompanying these processes. In order to visualize and efficiently demonstrate the construction of airport terminal building renovation and expansion, and to provide a rapid decision-making analysis tool for the control of structural safety risks, a digital twin model for safety risk prediction of steel mesh frame structures is proposed. On this basis, taking the construction situation of an airport as an example, sensors were installed according to the results of the analysis of the finite element model, and a prediction model was established using the height parameters of twelve hydraulic jacks, the ambient temperature data and the stress data, in order to predict the stress changes of the key rods of the steel mesh frame. The results show that the proposed prediction model can accurately capture the stress distribution of the steel mesh frame during the overall jacking process. The stress prediction accuracy can reach 98 % for the key bars. Through this method combining digital twin and machine learning, this study provides a new perspective for safety risk management in airport terminal building renovation and expansion projects, and verifies its applicability and effectiveness in actual projects through empirical studies.

1. Introduction

Airports play a vital role in the transportation system, and as the economy develops and the demand for domestic and international movement of people and businesses grows steadily, the actual throughput of existing airports has exceeded the design capacity at that time, and they need to be modernized and expanded [1]. Airport expansion projects usually include the construction of new terminals, the expansion of existing terminals and the construction of new runways and taxiways. With the increase in the number of large-span steel structure forms and the rapid expansion of the construction scale, large-span steel structure is a popular structural form for the construction of the terminal building. For the large-span steel structure, due to the huge and complex structural construction process, construction methods and construction techniques are complicated, the probability of risk in the construction phase is higher than other structures, while the construction industry is increasingly demanding for safety management intelligence. Moreover, once an accident occurs in a large and complex structure, its impact and consequences are extremely serious, causing huge property losses, casualties and serious social impacts [2–4]. Therefore, it is crucial to predict the structural safety risks and formulate corresponding safety measures in advance to reduce the structural safety risks during the construction process and safeguard the safety of construction personnel and the surrounding environment.

The structural safety risks of steel structures have been widely discussed and investigated in a number of past studies. For example, Rehman et al. [5] used STAAD.ProV8i to analyze and design a steel silo with the aim of determining the plate stresses, contour and critical supports, beam end force summaries, and rebar shedding in the silo structure. Tian et al. [6] established a finite element model of the structure during lifting, simulated the lifting process, and analyzed the effect of the displacement difference at the lifting points, proposed a new asynchronous analysis method called single and staggered points, and determined the optimal limit value of the displacement difference. Ruan et al. [7] proposed a construction method for sliding construction of replacement columns on large steel platforms with uneven elevations, which solved the construction challenges of steel sliding and column replacement on planes with different elevation angles. Yao et al. [8]

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make full use of the inherent elastic deformation range of steel structure to propose a prestressed construction process for small and medium-sized stadiums, which ensures the stability and integrity between the main truss and the roof.

All of the above studies are limited to the analysis of the overall simulation of the construction under certain conditions and do not take into account the safety performance of the structure at the next moment during the construction period. And based on the literature, statistics show that the traditional methods of predicting the stresses in various types of structures are summarized into three as follows (Table 1):

In recent years, some scholars believe that the traditional empirical methods, analytical methods, finite element analysis analysis methods can only obtain the mechanical parameters of the structure, can not obtain the key factors affecting the safety of the structure, and the prediction is low in precision and insufficiently intelligent. Machine learning techniques have been widely used in areas such as material property prediction [16,17] and image processing [18-20]. Some scholars have also carried out certain research in the field of construction engineering, real-time tracking of concrete vibration force through the relevant machine learning methods to realize the active control of concrete vibration quality [21]. The severity of damaged masonry buildings and their failure modes are assessed, among other things, by selecting appropriate crack characteristics and combining them with information about kinematics and static boundary conditions [22].

And with the rapid development of information technology, digital twin technology has become a key technical means to realize digital transformation and intelligent upgrading in the construction industry [23]. By integrating physical models, real-time data and analysis tools, digital twins map and simulate the behaviors and states of physical entities in virtual space, providing a new solution for monitoring and control of complex systems [24]. Although digital twins have been widely used in many fields such as smart manufacturing, smart city construction, healthcare, etc., the research in the field of safety of large-span steel structure construction is still in the exploratory stage.

Table 1
Traditional prediction methods for various types of structural stresses.

Methods	Features and Defects
FEA (Finite Element Analysis)	FEA is a numerical analysis method widely used in engineering to predict the stress, deformation and vibration of structures. By decomposing the structure into a finite number of cells and approximately resolving the stress and displacement fields within each cell, the behavior of the steel structure is simulated more accurately [6,9–11]. Involves simplifications and assumptions about structures, materials and boundary conditions, which may lead to some deviation of the analysis results from the actual situation, and for complex structures, it requires high performance computers, a large amount of memory and storage space, which leads to a longer time-consuming analysis process
NAMs(Numerical and Analytical Methods)	NAMs are usually based on theoretical equations and assumptions that simplify the structure and solve for analytical solutions. For example, for simple steel structures, methods such as beam theory or thin-wall structure theory can be used to predict the stress distribution [12–15]. NAMs may be difficult to apply directly to complex steel structure construction processes with many uncertainties and complex boundary conditions.
Experience and History	For specific types of steel structures, there also exist some empirical formulas and historical data for reference, such as the formulas in the construction code for steel structure engineering and the technical standards for the construction of steel mesh structure erection engineering. The safety status of the relevant structure can be obtained by analyzing the data. Empirical formulas and historical data references may have limitations as conditions may vary from project to project and need to be used with caution.

Currently, the existing studies mainly focus on the theoretical methods and framework construction of steel structure construction safety assessment. For example, Liu et al. [25] proposed an intelligent safety assessment method for prestressed steel structures based on digital twins, which improves the accuracy of construction safety assessment by fusing spatio-temporal multidimensional information. Zhu et al. [26] further combined digital twins with random forest algorithms to propose an intelligent analysis method for identifying key factors affecting the safety of prestressed steel structures. In addition, other studies have also combined digital twins with machine learning algorithms to improve the accuracy and real-time performance of structural safety assessment [27,28]. However, the structural forms are basically cable network structures, and the digital twins of experimental and simulated states are mainly considered, while there are few cases of studying the actual construction of large-span steel mesh frame structures at present.

This study is based on the construction process of the overall jacking of the steel roof grid structure of an airport renovation and expansion project, using digital twin technology, activating and passivating the cells through the finite element software Midas/Gen by means of the “life-and-death cell method”, and simulating the construction of the overall jacking phase of the solid grid, using the completed grid as the initial state of the jacking phase, and then simulating the overall jacking, installing sensors on the calculated key force members to collect data for the trial jacking and formal jacking, and then monitoring the construction process based on the actual data. The initial state of the jacking stage is the assembled mesh frame, and then the simulation of overall jacking is carried out. Sensors are installed on the calculated key force members to collect the data of trial jacking and formal jacking to realize the monitoring of the construction process, and then features are extracted based on the actual data, and the prediction models of GA-BP neural network, Random Forest and XGBoost are constructed, and the prediction accuracy and generalization ability of the three networks are compared and analyzed. Comprehensive use of multiple methods can improve the accuracy and reliability of the stress prediction of the steel mesh frame, and also provides a reference for future projects using the overall jacking method for construction.

2. Methodology

Fig. 1 illustrates the different parts of the proposed framework. The framework is divided into three main parts. First, the characteristics of structural safety risks are analyzed. The second part uses tools such as finite elements to build a digital twin model identical to the real construction project and data collection, and the third part builds a machine learning model to predict the safety risks during the construction of the terminal building.

The first part of this flowchart is to conduct a preliminary study to identify the various types of factors that may affect its structural safety risk and to explore the prediction strategy of structural safety risk based on the identified factors. The second part of the flowchart is to collect data from the construction through on-site fieldwork studies and with the help of various types of geometrical models created through BIM technology in the design phase to generate the corresponding finite element physical models and to arrange the sensors to form a digital twin model in the end. These fieldwork research studies aim to prepare a dataset of the actual state of the structural safety risk samples, whose specific characteristics have been characterized in the first part. In the third part, the dataset is utilized to build and train a machine learning model to learn the relationship between the features, and then the machine learning model is evaluated by appropriate performance metrics, and ultimately the structural safety risk is predicted using a superior predictive model.

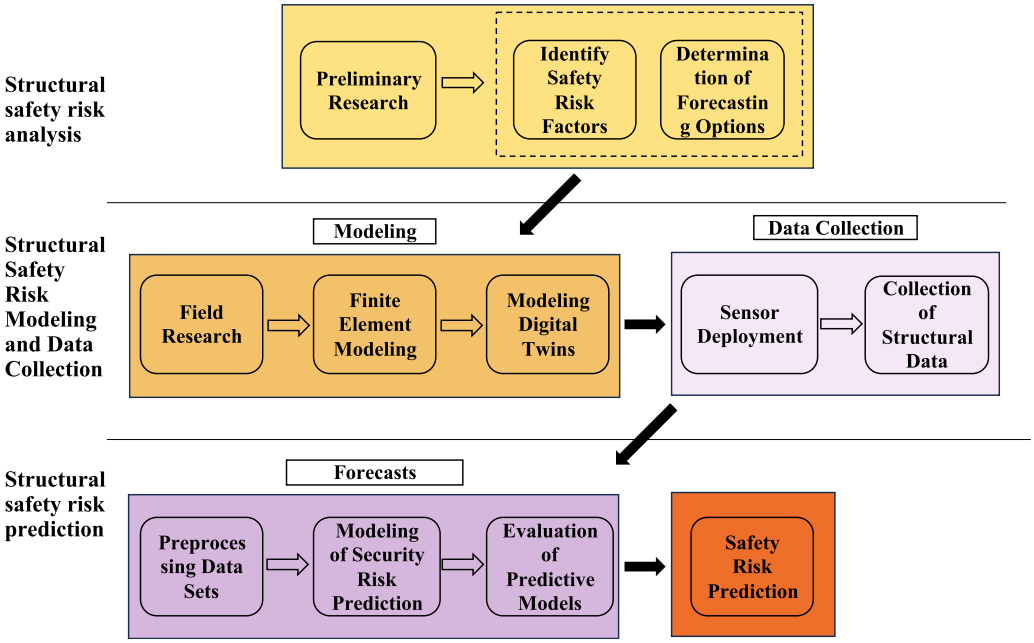


Fig. 1. Flow chart of the proposed methodology.

2.1. Digital twin framework

Digital twin technology is the key technology to realize the cross-fertilization of time and space dimensions, which can collect dynamic data from physical and virtual environments, realize the integration of physical information [29], and use machine learning algorithms as the driving force for real-time storage and processing of massive data [30,31].

Digital twin models enable one-to-one real-time simulation and mapping of structures under construction [23]. As shown in Fig. 2, a digital twin framework for construction risk prediction is established, i.e., modeling the geometric and physical dimensions of the solid structure, as well as the behavioral and rule dimensions that conform to the overall jacking construction scheme and related specifications, so as to realize the acquisition of structural information. Subsequently, through the real-time monitoring of sensors, the massive data is acquired and a prediction model is built to realize the prediction of structural safety risks, and ultimately realize the control and guidance of the overall construction.

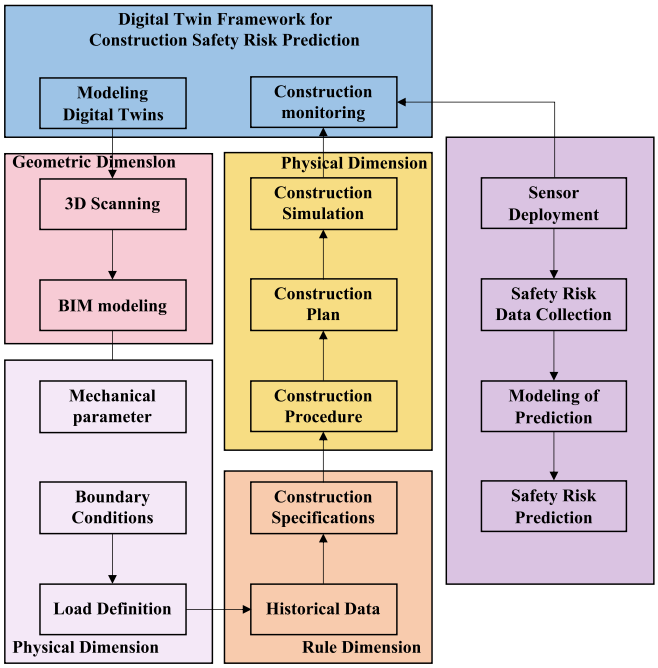


Fig. 2. Digital twin framework for construction safety risk prediction.

2.2. Correlation prediction algorithms

Through the actual project and Midas finite element software analysis, it can be seen that the stresses and displacements in the key parts of the steel mesh roof cover using the overall jacking method under the time-varying conditions are affected by the height of the jacking frame. The analytical method and the method of empirical formulas and historical data are suitable for relatively simple structures, while the finite element method cannot predict the structural stress and displacement at the next moment in real time, which is obviously not fully applicable for the need to pay attention to the safety risk of the steel mesh frame during the construction of the overall jacking, so it is necessary to realize the monitoring and prediction of the structural safety risk through the algorithm based on machine learning. Most of the previous researchers use RF (Random Forest) algorithm and BPNN (Backpropagation neural network) when they use machine learning methods to study the safety problems related to steel structure, and this study is the first time to use XGBoost (Extreme Gradient Boosting) algorithm to construct the model.

2.2.1. BPNN

BPNN is a common artificial neural network model for supervised learning. It continuously adjusts the weights in the network through a back propagation algorithm to minimize the error between the output and the target. BP neural network usually consists of an input layer, a hidden layer, and an output layer, where each neuron is connected to all the neurons in the previous layer and has weights and biases. During training, the weights and biases are updated by a back propagation algorithm to make the output of the network as close as possible to the true value. However, the loss function of traditional BP neural networks is usually non-convex, such as easy to fall into local optimum and slow convergence. To overcome these problems, researchers often use optimization algorithms such as GA (Genetic Algorithm) and PSO (Particle Swarm Optimization) to improve BPNN. Genetic algorithm is an

optimization algorithm that simulates the process of biological evolution and searches for optimal solutions by simulating genetic operations such as natural selection, crossover and mutation. A wider solution space can be searched through the diversity and randomness of the population, which helps to avoid falling into local optimal solutions. Optimizing the weights and structure of the neural network through genetic algorithms can make it more likely to find the global optimal solution [32].

2.2.2. RF

RF is an integrated learning algorithm that improves the accuracy and stability of predictions by constructing a combination of multiple decision trees. During the construction of RF, each tree is trained independently based on a randomly selected subset of features and randomly sampled samples, and this randomness reduces overfitting, increases the generalization ability of the model, and also makes the whole forest more robust. Each decision tree makes a classification or regression prediction on its input samples, and then RF arrives at the final prediction by integrating the predictions of all trees and voting or averaging them. This integration strategy helps to reduce the errors that may arise from a single decision tree and improves the stability and accuracy of the overall prediction. RF can be used not only for classification problems, but also for regression problems, and usually exhibits good performance. RF also has the advantages of being easy to implement, insensitive to feature selection, and capable of evaluating feature importance. Therefore, RF has gained much attention in practical applications [33,34].

2.2.3. XGBoost

XGBoost is an efficient and scalable gradient boosting decision tree algorithm that optimizes the loss function and model complexity to minimize the prediction error and control overfitting. In XGBoost, a new decision tree is added in each iteration to fit the residuals of the previous round, and the true value is continuously approximated by accumulating the predictions of all trees. In order to improve the efficiency, XGBoost uses various optimization strategies such as column sampling, regularization, and second-order derivative information to accelerate the optimization process. In addition, XGBoost supports parallelized computation, which can make full use of multi-core CPUs or distributed computing resources to further improve the training speed. These features enable XGBoost to perform well on large-scale datasets, and it has achieved excellent results in many machine learning competitions and real-world applications [35,36].

2.3. Evaluation indicators

The prediction accuracy of the model in this study is mainly measured by the Mean Squared Error (MSE), Mean Absolute Error (MAE), Mean Bias Error (MBE) and the Coefficient of Determination (R^2), which are calculated as follows: The prediction accuracy of the model in this study is mainly measured by the mean square error (MSE), mean absolute error (MAE), mean bias error (MBE) and the coefficient of determination (R^2), which are calculated as shown in Eqs. (1)–(4):

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{1}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{2}$$

$$MBE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i) \tag{3}$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y}_i)} \tag{4}$$

Where: y_i is the actual value; $\hat{y}_i$ is the predicted value; $\bar{y}_i$ is the mean value; n is the number of samples; *SSE* is the sum of squares of residuals; *SST* is the sum of squares of the total deviation. The smaller the *MSE*, *MAE*, and *MBE* are, the closer the fitted results are to the actual values, i.e., the better the model prediction effect. The R^2 indicates the goodness-of-fit of the function, with a value in the range of [0,1], and the closer it is to 1, the better the function is fitted.

3. Digital twin model development

3.1. Project overview

This study takes an airport renovation and expansion project as an example, as the airport has exceeded its design capacity, in order to improve the safety capacity of the airport terminal area, enhance the level and quality of airport services, and adapt to the needs of the growth of the airport's aviation business volume. At the same time, it is conducive to the creation of local three-dimensional transportation, promote industrial agglomeration and development, and drive urban economic growth. Reconstruction and expansion of the airport project, the main structure of the project for the reinforced concrete frame structure, roof using steel mesh frame structure, transformation of the original terminal building 998m^2, the new terminal building construction area of 26002m^2, the largest overhang of about 12 m, the net frame as a whole adopts positively placed quadrangular cone net frame structure, node bolted ball, welded ball connection, part of the bolted ball connection, the maximum span of 35.15 m, the maximum overhang of 12.56 m, the overall thickness of the net frame is 2.0 m.

During its construction, the incompleteness of the structure, the mobility of the constraints, and the complexity of the stress conduction overall lead to a greater safety risk of the structure during the construction phase [25], and in response to the current high-precision, real-time requirements for the prediction of construction safety risk, prediction through structural health monitoring is no longer sufficient to meet the demand [36]. Therefore, based on the steel mesh frame structure of an airport renovation and expansion project, the digital twin concept is used as a driver for the prediction of structural safety risks under spatial and temporal change scenarios for structural safety risk elements, so as to ensure the safety of construction.

3.2. Structural modeling

3.2.1. FEA modeling

First, the structural model of the airport terminal building reconstruction and expansion project is established. This study establishes a fine geometric model through 3D laser scanning technology and BIM technology, imports the geometric model into Midas finite element software, and establishes the corresponding finite element model physical model by taking into account the geometry, material properties, boundary conditions and other factors of each part. The jacking

Table 2
Mesh frame finite element details and description.

Details	Descriptions
Materials	Q235B, Q355B
Poisson Ratio (steel)	0.3
Elastic modulus	2.06 × 105 N/mm2
Coefficient of linear expansion	1.2 × 10–5 1/°C
Specific Weight (steel)	7.698 × 10–5 N/mm3
Jacking frame cross-section characteristics	P 60 × 3.5, P 159 × 6, L 50 × 5, HW 200x204x12/12, P 219 × 10
Steel mesh frame cross-section characteristics	P 75.5 × 3.75, P 88.5 × 4, P 114 × 4, P 140 × 4, P 140 × 6, P 159 × 8, P 159 × 12, P 180 × 12, P 219 × 14, P 245 × 14, P 273 × 16, P 299 × 16
Nature of the element	Beam element

equipment uses DS-100C-1250 hydraulic jacking jacks, and the materials used and detailed information are shown in Table 2 Since this study only carries out the study of the jacking stage, without considering the assembly process before jacking and the unloading process after jacking, so only the jacking stage of the mesh frame and jacking frame model is established, a total of 1813 nodes and 6152 units are established, as shown in Fig. 3.

The jacking points should be evenly arranged under the net frame to ensure the synchronization of jacking, which is the key to the success of jacking operation. Multiple jacking points should be set up horizontally in a single row in each jacking area, and the jacking points should fully consider the geometrical plane size of the distribution of the mesh frame, and be set up under the bolt ball of larger diameter as far as possible, so as to ensure the overall stability of the mesh frame in the process of jacking. And through certain calculation and analysis, use the space rack structure analysis system to check the internal force of the rack and the force of the jacking rack during the jacking construction to ensure the structural safety, the position of the jacking rack is shown in Fig. 4.

3.2.2. *Integral jacking simulation*

According to the needs of the reconstruction and expansion project, the overall jacking construction plan is discussed and designed, including the assembly sequence, the selection of jacking frame and the arrangement of jacking points. The whole process of finite element simulation is carried out by Midas/Gen, and according to the local structural changes and boundary condition changes during the jacking process, the activation and passivation operations are carried out on the relevant contents using the method of living and dead units, and the mass matrix and stiffness matrix of the corresponding units are zeroed out, and the corresponding stiffness matrix is multiplied by 1×10^6 so that its action is very small without singular values. The mass matrix and stiffness matrix of the corresponding unit are zeroed, and the passivated unit load is 0. The corresponding stiffness matrix is multiplied by 1×10^{-6} so that its role is small and it does not produce singular values. After the cell is activated, the mass and stiffness matrices are restored to their original values and the overall calculation is performed.

The area of the net frame in the jacking area of this project is about 5350 m2 with a weight of 360.1 t and a difference in jacking height of 14 m. The total weight of the net frame is about 360.1 t and the load sub-coefficient is 1.4. During the lifting process, there is a certain degree of height deviation when the various jacking points of the roof are rising synchronously. Therefore, it is necessary to consider the effect of asynchronous lifting points on the structure during the lifting process. Asynchrony mainly affects the stresses in the members around the lifting points. Based on engineering experience, a power factor of 1.2 times is considered. Therefore, the total coefficient for checking the strength of the bars is $1.4 \times 1.2 = 1.68$, and the sub-coefficient for checking the deformation of the mesh is 1.0. The time of the jacking operation is about 5 days, and the weather factor is within the controllable range,

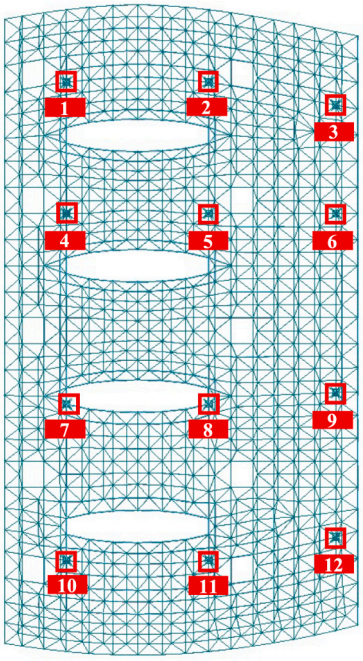
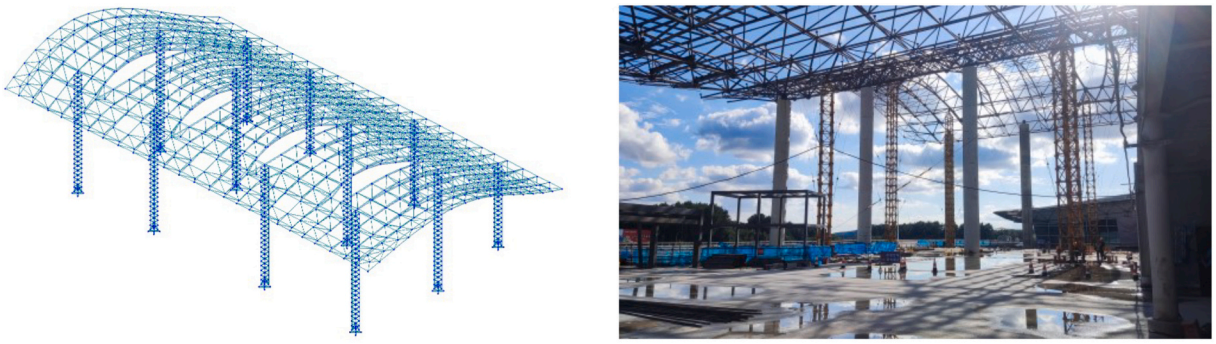


Fig. 4. Schematic arrangement of jacking frame.

and the roof is a hollow structure, which is subjected to small wind loads. Therefore, the effect of wind load is not considered for the time being when analyzing the lifting conditions. Under the overall lifting conditions, temperature loads are not considered. And in accordance with the overall construction process, each lifting 1 m is calculated as a construction step, and a total of 15 construction steps are divided into molding, lifting 1 m ~ lifting 14 m (Fig. 5).

Replacement of stress-exceeding bars and structural reinforcement during construction suspension. Through the finite element analysis results, the stress and displacement data of the key points in the jacking process were obtained. The maximum tensile stress of the rods in the jacking stage was 184.3 MPa and the maximum compressive stress was −142.2 MPa, both of which appeared in the rods connected to the support apex of the jacking frame during the jacking construction. Specific details are shown in Table 3. The stress sensors were arranged according to the location of the maximum stress to monitor the overall jacking process in real time (Fig. 6).

Arrangement of stress sensors according to the location of the maximum stress value, real-time monitoring of the overall jacking process.



(a) Finite element model (b) Physical entity

Fig. 3. Integral jacking steel mesh.

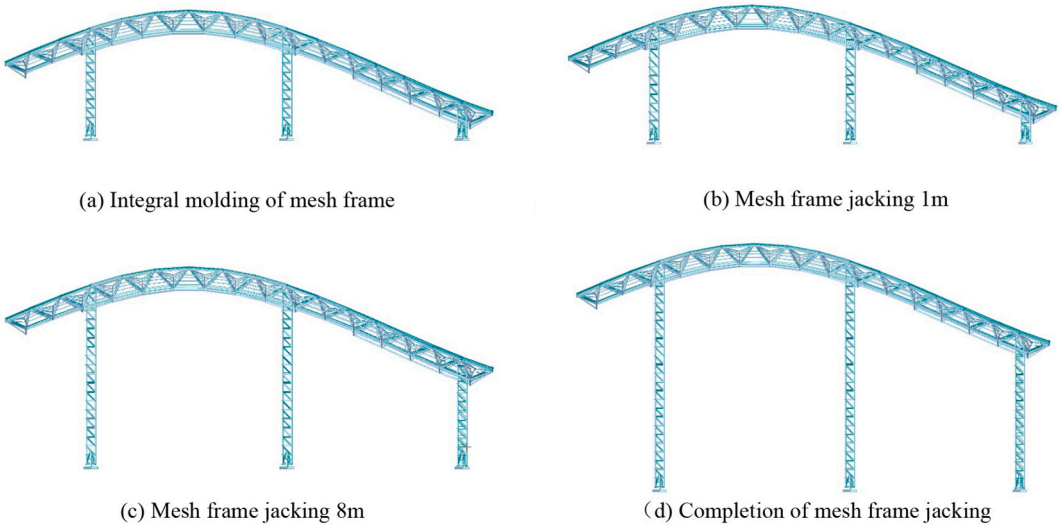


Fig. 5. Overall jacking process.

Table 3
Simulation results of mesh rack jacking process analysis results.

Construction Stage	Max Stress/MPa	Min Stress/MPa	x-Displacement/mm	y-Displacement/mm	z-Displacement/mm
Jacking 1m	184.2	−139.5	−7.8	3.9	−21.1
Jacking2m	184.3	−139.0	−7.7	4.1	−21.2
Jacking3m	184.0	−138.5	−7.6	4.1	−21.2
Jacking4m	184.0	−138.1	−7.4	4.1	−21.2
Jacking5m	184.3	−137.9	−7.2	4.0	−21.2
Jacking6m	184.2	−137.7	−7.1	4.0	−21.2
Jacking7m	184.2	−138.6	−7.0	3.9	−21.2
Jacking8m	184.2	−138.9	−6.9	3.8	−21.2
Jacking9m	184.3	−139.1	−6.8	3.7	−21.2
Jacking10m	184.0	−139.3	−6.7	3.6	−21.2
Jacking11m	184.2	−139.4	−6.6	3.6	−21.2
Jacking12m	184.2	−139.6	−6.5	3.5	−21.2
Jacking13m	184.2	−139.7	−6.4	3.4	−21.2
Jacking14m	184.2	−142.2	−6.3	3.4	−21.2

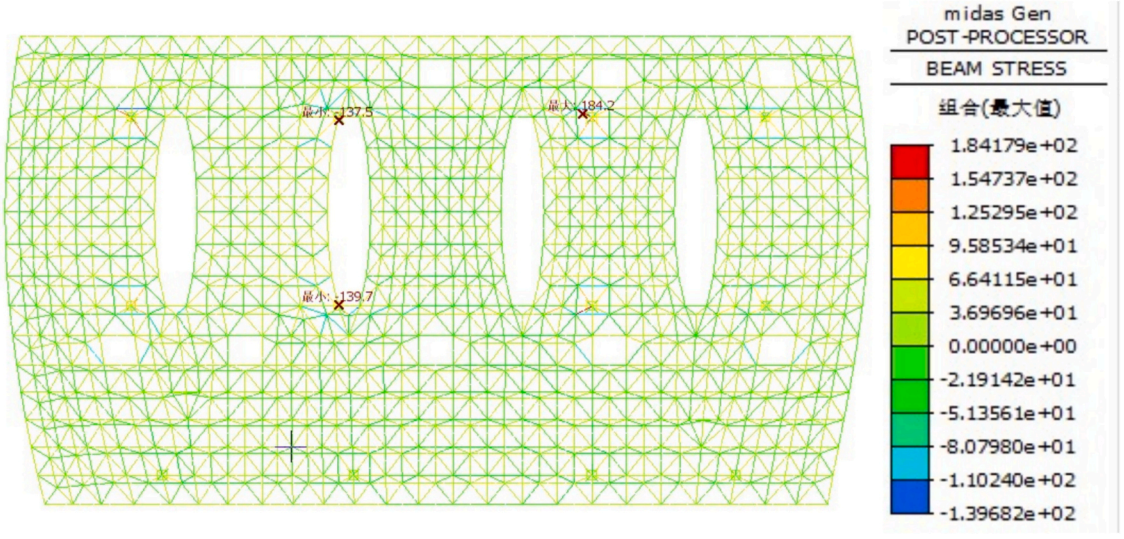


Fig. 6. Finite element simulation results.

3.3. Digital twin model

Driven by the digital twin technology, for the current limited construction budget, to ensure the maximum effectiveness of the sensor monitoring, but also to ensure that the diversity and accuracy of the acquisition of the data of the key rods of the solid structure, for the construction of the subsequent predictive model database to provide a reliable guarantee. In addition, the established high-fidelity digital twin

model itself is a mapping of the solid structure, which can complete the simulation of the actual construction, and can realize the construction under all kinds of working conditions, and can provide the damage data and over-limit data that can not be detected by the actual construction, so as to realize the completeness of the prediction database. As shown in the Fig. 7.

4. Prediction models

4.1. Data acquisition and processing

In the actual construction process, the stress of key rods, the height of hydraulic jacks, the ambient temperature and other data are collected through sensors and hydraulic synchronization computer control system and other devices. The sensor and the corresponding jacking system are shown in Fig. 8.

To establish the measured data set corresponding to the finite element model data, it is necessary to process and analyze the collected actual data, extract the key information, and combine it with the simulation data. The actual data collected at the construction site are generally characterized by high dimensionality, high noise, and serious covariance, and need to be pre-processed, including data screening, anomaly processing, and data noise reduction. And when necessary, the data need to be standardized to eliminate the effect of orders of magnitude and units. Because in the use of machine learning algorithms, if you do not remove the independent variables with a significant linear relationship, it will cause the model performance degradation.

4.1.1. Training sample

Measured data may be missing or abnormal due to sensor performance and external factors [37], and processing of missing and abnormal data is an important prerequisite to ensure the effectiveness of model prediction. Each sensor works independently and does not interact with each other, but the detected temperature data may be slightly fluctuating due to direct sunlight or being shaded by rods, so the temperature data is averaged as a feature of the input data. At the same time, the height of the twelve jacking frames is also used as a feature of the output data, which together with the temperature data constitutes the input data, and the structural stress data monitored by each sensor constitutes the output feature of the data (Table 4).

In order to analyze and study the structural safety risk of steel mesh frame roof cover, this study selects one of the most representative datasets for experiments. The monitoring data are the jacking height data from the beginning of jacking to the jacking pause with the stress data of the corresponding steel mesh frame key rods, and the sensor monitoring data are collected according to the jacking construction date of the project (10/3, 10/4, 10/6, and 10/7, four days in total).

First, the data set is screened to extract the jacking frame monitoring data, stress sensor monitoring data, and temperature sensor monitoring data of the site, and a total of 500 sets of valid data are obtained, and due to the large amount of data, only some sets of corresponding data are shown in this study as shown in Appendix Table A1 and A2. Missing value processing is carried out for each feature of input data and output data in turn, and the processing strategy is selected according to the missing proportion p. Linear interpolation is utilized to make up the missing values when the missing proportion is lower than p. After that, outliers are processed, and outliers usually refer to observations that deviate far from the whole sample as a whole. Detect outliers according to the data distribution, calculate the mean μ and standard deviation σ of each data feature respectively, and the observed values are considered outliers when they are larger than $\mu + 3\sigma$ or smaller than $\mu-3\sigma$. Linear interpolation filling method is used to deal with the outliers, i.e., linear interpolation is performed on the data adjacent before and after the outliers, and the interpolation results are used to replace the outliers.

After a thorough cleaning of the data, including dealing with missing values and outliers, we performed data noise reduction. For the jacking frame data, we observed relatively smooth changes in the values with a limited range of fluctuations, so we decided not to perform additional noise reduction on it. This decision was based on a careful analysis of the data characteristics to ensure the efficiency and accuracy of the subsequent analysis. In contrast, the data from the stress sensors showed significant fluctuations, which could have an impact on the accuracy of structural health monitoring. To address this issue, a classical data smoothing technique, the sliding mean filtering algorithm, is used. This algorithm reduces the effect of random noise by creating a sliding window of size m and averaging the data points within the window. In

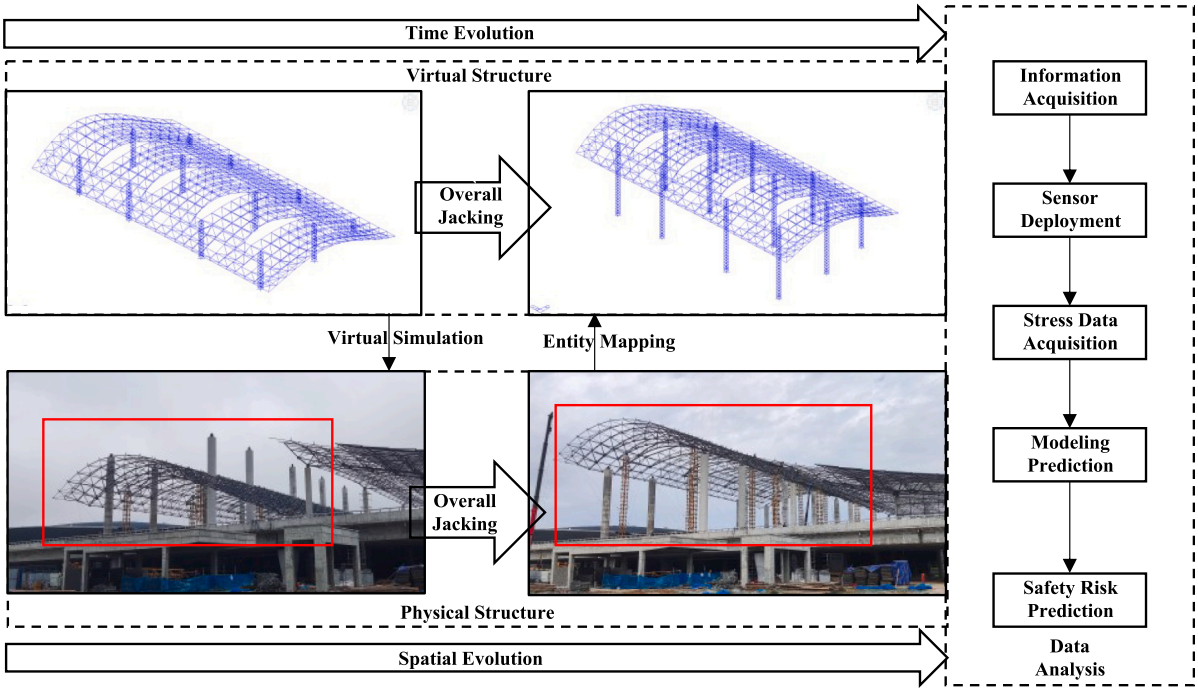


Fig. 7. Digital twin framework for safety risk prediction considering spatio-temporal variations.



(a) Sensor mounted on element (b) Synchronized jacking control system

Fig. 8. Data collection devices.

Table 4
Characterization of the dataset.

Feature	Descriptions	Unit
Jacking height	Height of hydraulic jacks during jacking process	mm
environmental temperature	Real-time monitoring data of ambient temperature during construction	°C
Stresses	Stress monitoring data for critical bars of steel mesh frames	N/mm^2

this case, we carefully chose m = 10 as the size of the sliding window, a value that has been shown in experiments to be effective in balancing the noise reduction effect with the retention of data details. In Fig. 9, the significant effect of the noise reduction treatment can be clearly seen. The curve on the left side represents the critical rod stress data monitored by the sensor before noise reduction, in which the fluctuation of the data points appears to be more disorganized, which may be due to a variety of factors, such as transient errors of the sensor, small changes in the environment, and so on. The curve on the right, on the other hand, shows the data after noise reduction, where the data points have been converted into a more continuous and smooth curve, better reflecting the stress changes of the structure over time. This continuity not only makes the data easier to analyze, but also reveals the actual behavior of the structure more accurately, providing more reliable data support for structural health monitoring.

Finally, the jacking frame height data are matched with the structural stress data, and the measured moments of the jacking frame height data are set to correspond to the structural stress data in order to realize the safety risk prediction of structural stress. After the data matching is completed, the full set of samples contains 488 sets of data, totaling 13,664 data samples. Due to the large amount of data, only part of the corresponding data sets are shown in this study as shown in Appendix Table A1 and A2.

The data after preprocessing are presented as follows. The data presented in the table are the data obtained from the first 10 acquisitions on 10/3, when the initial overall jacking construction was carried out. The "Serial Number" in the first column of Appendix Table A1 is used to arrange the time sequence of the data, and the first row of the table is the height of the jacking and the serial number of the corresponding jacking frame, and the corresponding position of the serial number is shown in the previous figure. The serial number in the first column of Appendix Table A2 is the same as that shown in the table, and the serial number shown in the first row is the corresponding serial number of the sensor.

4.1.2. *Input and output data preprocessing*

Next, in order to improve the effectiveness of the neural network, the dataset arrangement was randomly disrupted to exclude the learning piecemeal of the original data due to temporal and spatial aggregation. The dataset is divided into 2 parts according to 9:1, and the first 440 groups of reordered data are selected as the training set, and 48 groups of data are selected as the validation set. In this way, 15 rounds are executed, 15 model evaluation results are obtained, and the average of the 15 evaluation results is taken as the overall performance of the algorithm.

Considering the different scales of input and output quantities, and in order to improve the accuracy of the prediction model, the scale trans-

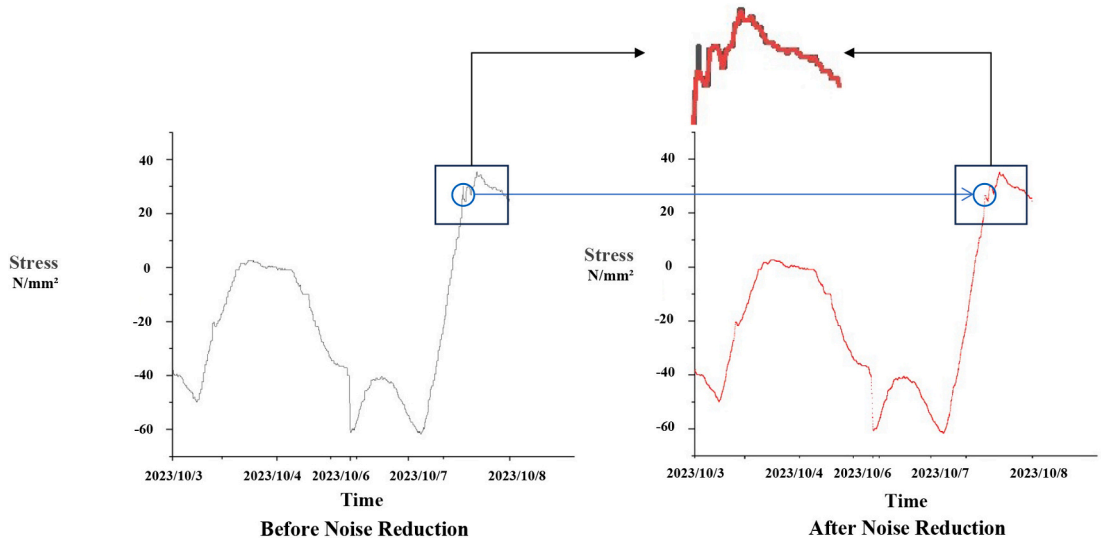


Fig. 9. Effect of noise reduction for height difference data of some stress sensors.

formation of the experimental data before the model is trained requires standardization of the parameter data, which is usually normalized by restricting the input and output data to the interval of [0,1] with the formula *Z*-Score Normalization is used in this study, and the eigenvalues into a distribution with mean 0 and standard deviation 1, as shown in Eq. (5).

$$Z = \frac{X - \mu}{\sigma} \tag{5}$$

where: where μ is the mean of the feature and σ is the standard deviation of the feature.

4.2. Predictive modeling

In this study, a structural safety risk prediction model based on GA-BPNN, RF, and XGBoost was built using Python.

4.2.1. GA-BP

Hidden layer neurons (N3) and population size (G) are important parameters affecting the performance of GA-BP neural networks, however, no generalized formula has been found to determine these two parameters. In order to guarantee the predictive performance and fault tolerance of the model, it is necessary to determine the appropriate parameters of the GA-BP model through test selection. In the GA-BP neural network, the fitness function first utilizes data to train the BP neural network and takes the prediction error as the individual fitness value. Then after a series of selection, crossover, mutation and repetition operations, the optimal individual corresponding to the optimal fitness value is finally found to improve the accuracy of the model. The hidden layer neurons and population size are important parameters affecting the performance of the GA-BP neural network, in order to guarantee the model prediction performance and fault tolerance, the following empirical formula is used to calculate the rough estimation of the number of nodes of the hidden layer in the range of 3–13, and the optimal number of nodes of the hidden layer is 13 and the population size is 150 using the trial-and-error method, as shown in Eqs. (6)–(8).

$$h = \sqrt{m + n} + a \tag{6}$$

$$h = log_2 m \tag{7}$$

$$h < m - 1 \tag{8}$$

Where: h is the number of neuron nodes in the hidden layer; m is the number of neuron nodes in the input layer; n is the number of nodes in the output layer; a is usually a constant between 1 and 10.

4.2.2. RF and XGBoost

The parameters of the prediction model based on the RF algorithm and the XGBoost framework are shown in the Table 5, and the default values of the algorithm are adopted except for the number of base learners, in which the maximum tree depth is “None” to indicate that there is no restriction on the maximum tree depth.

5. Results and discussion

The established prediction model of the degree of inclination of the sinkhole is tested using the completed preprocessed sensor monitoring data. The prediction model adopts GA-BP neural network, RF algorithm and XGBoost framework, respectively. After each round of training and testing, the predicted stress values of key structural rods are output, and the predicted values are compared with the corresponding measured data. In order to visualize the prediction effect, the discounted graphs comparing the predicted values with the actual values and the model fitting effect graphs are drawn, with the first line being the GA-BPNN-based prediction model, the second line being the RF-based prediction model, and the third line being the XGBoost-based prediction model. For when the actual value is closer to the predicted value, the deviation of the corresponding line graph is smaller, and the fitting effect graph is closer to the dotted line in the figure. From the last line of the picture, it can be seen that except for some points sporadically distributed in the region farther away from the dotted line, the rest of the points are centrally distributed near the dotted line, indicating that the predicted value is very close to the actual value, and the consistency between the two is good, and the model achieves better prediction results. As shown in Fig. 10.

In order to make the prediction effect more significant, quantitative indexes are used to evaluate the prediction effect of the above three models, MSE, MAE, MBE and R2, which are calculated as shown in the Table 6. The table shows that the RF and XGBoost models perform similarly, and the goodness-of-fit R2 is significantly better than that of the GA-BPNN model. Although the validation set R2 of the RF model is closer to the training set R2, the MAE value is larger and R2 is smaller, so there is a certain degree of overfitting, while the XGBoost model has no obvious overfitting or underfitting.

The MSE values of the two models for structural stress prediction are less than 5, the MAE values are less than 0.2, the MBE values are less than 0.1, and the R2 values are more than 0.9. Combined with the mean values of the evaluation indexes, it can be seen that the prediction accuracies of the models based on the RF algorithm are slightly lower than that of the model based on the XGBoost, but the differences in the prediction accuracies of the two models are relatively small, and the prediction results are sufficient to meet the needs of the project.

Further, given the total sample size of this study, the use of GA-BPNN calculations would result in poor accuracy. The intention of this study is to predict the stresses on critical bars during construction and to use a small amount of data during trial jacking for training, which requires a more comprehensive use of data in constructing the model. XGBoost comes with cross-validation during training, which excludes the problem of model overfitting. Therefore, the XGBoost algorithm chosen in this study is more effective in constructing the model.

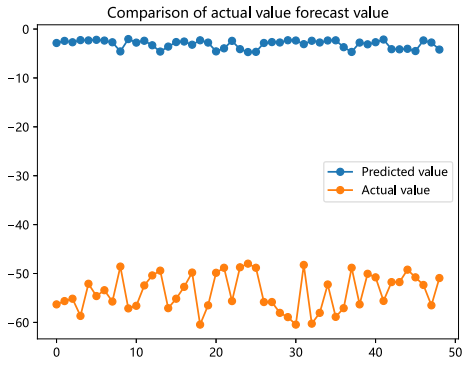
6. Conclusions

In this paper, a structural safety risk prediction method for airport terminal building construction with comprehensive consideration of spatial and temporal changes is proposed, focusing on the overall jacking process of steel mesh frame roof. The prediction of stresses in key parts of the steel mesh frame is realized through the steps of establishing a structural model, simulating the overall jacking construction, arranging and installing sensors, data acquisition and processing, and establishing a stress prediction model. This method provides a new idea and method for the safety management of the terminal building construction project, which has certain theoretical and practical significance The main research conclusions and prospects are as follows:

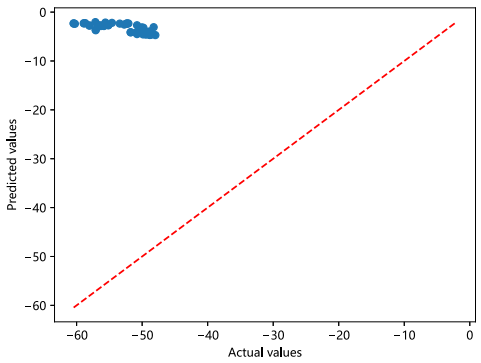
The performance of three machine learning prediction models, GA-BPNN, RF and XGBoost, is compared, and the XGBoost model has the

Table 5
Parameterization of the prediction model.

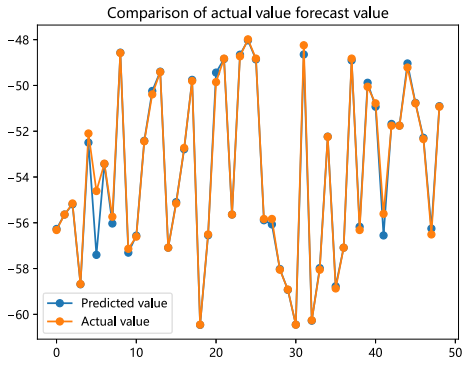
	Maximum tree depth	Minimum number of samples required for splitting	Minimum number of samples for leaf nodes	Number of base learners	L1 regularization parameter	L2 regularization parameter	Learning rate
RF	None	2	1	1000	–	–	–
XGBoost	None	–	–	1000	0	1	0.1



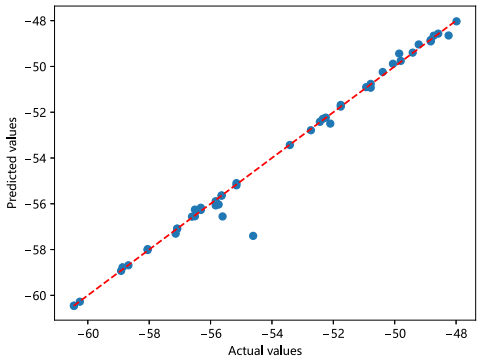
(a) Comparison of actual and predicted Ga-bp values



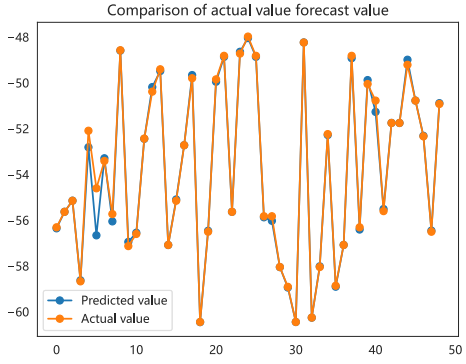
(b) Ga-bp model fitting effect



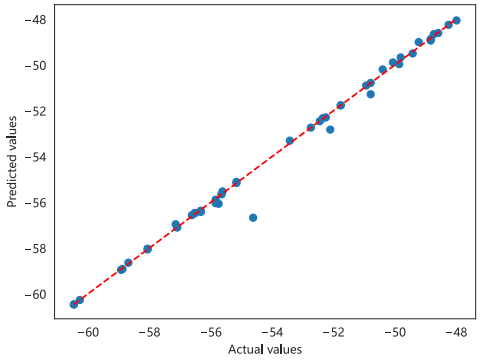
(c) Comparison of actual and projected RF values



(d) RF model fitting effects



(e) Comparison of actual and predicted XGBoost values



(f) XGBoost model fitting effects

Fig. 10. Comparison of partial stress predictions.

Table 6
Model effect.

	GA-BPNN		RF		XGBoost	
	Training set	Validation set	Training set	Validation set	Training set	Validation set
MSE	2590.652	2559.307	0.0129	0.1856	3.6452	0.1101
MAE	50.795	50.5055	0.0415	0.1543	0.0004	0.1248
MBE	−50.795	−50.5055	0.0031	0.0789	2.2975	0.0618
R^2	−155.588	−185.877	0.9992	0.9864	0.9999	0.9919

best performance in terms of model performance evaluation, prediction accuracy, and operation time. The stress prediction results are basically consistent with the stresses of the key rods during the overall jacking period, and the stress prediction model of the key rods has R2 of 0.9919 and MBE of 0.0618, which are able to provide more reliable stress values.

The prediction model based on two different algorithms of machine learning can reasonably predict the structural safety risk of the steel mesh frame during the overall jacking process. Combined with the real-time monitoring data of the jacking height of the jacking frame during the overall jacking period, it can accurately predict the stresses of the key rods of the steel mesh frame, which can provide strong technical support to ensure the safety of the construction structure.

In this paper, the real monitoring value of the construction site is used as the machine learning training set, which has the problem of error superposition. With the development of intelligent perception technology and the improvement of machine learning accuracy, in the future, we can also use the height sensor of the jacking frame combined with the temperature sensor and the wind speed and direction sensor to input real-time data, combined with the data assimilation algorithm prediction model, to synchronously improve the real-time prediction and accuracy.

In order to further enhance the application value and scientific contribution of this research, future work will focus on the following key directions:

1. Multi-source data fusion and integration with real-time monitoring systems: We plan to explore how to integrate data from multiple sensors, including environmental parameters and structural responses, to improve the accuracy and robustness of predictive models. In addition, we investigate how to integrate these models with real-time monitoring systems for automated construction safety warnings, with a view to providing more timely safety risk assessments in practice.
2. Algorithm optimization and model generalization capability: Future research will be devoted to optimizing existing machine learning algorithms and adjusting parameters to improve prediction accuracy and computational efficiency. At the same time, we will also evaluate the generalization ability of the model under different types of steel structures and construction conditions, with the aim of developing a more general safety risk prediction tool.

Through these directions, we expect to provide new theoretical insights and practical tools in the field of safety risk prediction for steel structure construction.

CRediT authorship contribution statement

Zhansheng Liu: Writing – review & editing, Project administration, Investigation, Funding acquisition. **Chengkuan Ji:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Guoliang Shi:** Formal analysis, Visualization, Writing – review & editing. **Yanchi Mo:** Formal analysis, Visualization, Writing – review & editing.

Declaration of competing interest

The author declares no conflict of interest.

Appendix A. Appendix

Appendix Table A1
Elevator height data.

Jacking frame Height (mm)	Height 1	Height 2	Height 3	Height 4	Height 5	Height 6	Height 7	Height 8	Height 9	Height 10	Height 11	Height 12
1	17.12	17.56	19.04	18.12	18.39	18.21	17.59	18.14	17.71	18.56	17.26	18.41
2	35.27	35.95	35.54	35.40	37.02	36.17	36.03	35.66	36.54	36.00	35.32	36.55
3	41.93	43.01	42.38	42.71	42.36	42.79	43.56	43.42	42.72	42.36	42.04	41.98
4	51.75	53.65	52.96	52.80	53.45	53.73	52.12	52.20	53.60	51.82	52.43	53.30
5	79.46	80.32	79.68	81.01	81.42	81.43	81.24	79.61	79.48	81.44	81.22	79.79
6	92.04	92.30	93.55	93.13	92.61	92.68	93.24	93.71	93.92	93.77	92.17	93.79
7	134.01	134.36	134.59	135.96	135.28	135.07	135.82	134.12	135.78	135.49	134.56	134.27
8	151.06	152.83	152.52	151.09	152.49	152.22	151.41	153.06	152.24	152.00	152.36	151.97
9	168.16	169.07	169.24	168.90	170.06	168.42	169.53	169.47	169.05	169.91	169.37	168.64
10	177.04	177.61	178.52	177.87	177.39	178.89	177.53	177.58	177.72	178.08	177.70	178.94
…	…	…	…	…	…	…	…	…	…	…	…	…

Note: The data listed in this table are in m.

Appendix Table A2
Stress data.

Sensors Press (MPa)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	49.66	31.65	54.82	26.56	151.63	136.12	80.96	58.69	57.40	53.69	58.69	39.26	101.66	54.70	52.25
2	49.57	31.55	54.73	26.61	151.58	136.18	80.87	58.73	57.40	53.64	58.73	39.30	101.66	54.61	52.25
3	49.47	31.45	54.63	26.67	151.54	136.23	80.78	58.78	57.40	53.59	58.78	39.35	101.66	54.46	52.10
4	49.38	31.36	54.54	26.72	151.50	136.29	80.69	58.83	57.40	53.54	58.83	39.35	101.48	54.32	51.95
5	49.28	31.26	54.45	26.77	151.50	136.29	80.60	58.88	57.40	53.50	58.88	39.21	101.30	54.17	51.81
6	49.19	31.16	54.36	26.82	151.50	136.47	80.51	58.88	57.40	53.45	58.88	39.07	101.12	54.03	51.66
7	49.10	31.06	54.36	26.88	151.46	136.65	80.42	58.88	57.40	53.40	58.88	38.93	100.94	53.88	51.51
8	49.00	30.97	54.36	26.88	151.42	136.82	80.34	58.73	57.20	53.36	58.73	38.79	100.76	53.74	51.37
9	49.00	30.87	54.22	26.88	151.38	137.00	80.25	58.59	57.01	53.31	58.59	38.66	100.58	53.59	51.22
10	49.00	30.87	54.08	27.14	151.33	137.18	80.16	58.45	56.81	53.26	58.45	38.52	100.40	53.45	51.07
…	…	…	…	…	…	…	…	…	…	…	…	…	…	…	…

Note: The data presented are compressive stresses in MPa, except for the fourth and sixth columns, which are tensile stresses.

Data availability

Data will be made available on request.

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